Ofqual QAN	**600/0406/X, 600/0624/9**
Quartz ID	**1156, 1157, 1060**
Qualification	**Level 4 Diploma in Computing, Level 4 Diploma in Computing (with Business Management), Level 4 Diploma in Business Information Technology**

Assessment	**Skills for Computing**
Assessment Type	**Global Examination**
Date	**Wednesday 5th March 2025**
Time	**09:00AM-12:00PM**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

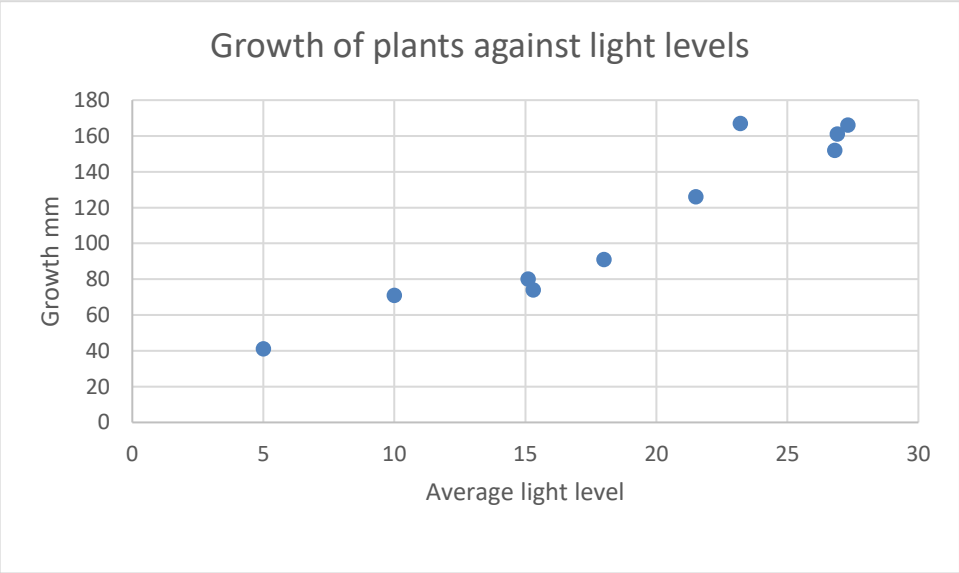
Question 1

A horticulturist measures the amount of light given to TEN (10) specimens of the same plant over a year. The average light each plant receives and the growth of each plant in mm over the year is recorded in the table.

Plant	Average light level	Growth (mm)
1	10	71
2	15.1	80
3	23.2	167
4	26.9	161
5	27.3	166
6	18	91
7	15.3	74
8	26.8	152
9	5	41
10	21.5	126

a) Draw a scatter graph to show the relationship between the average light level and the growth over the year. **7**

Mark scheme:



1 mark each

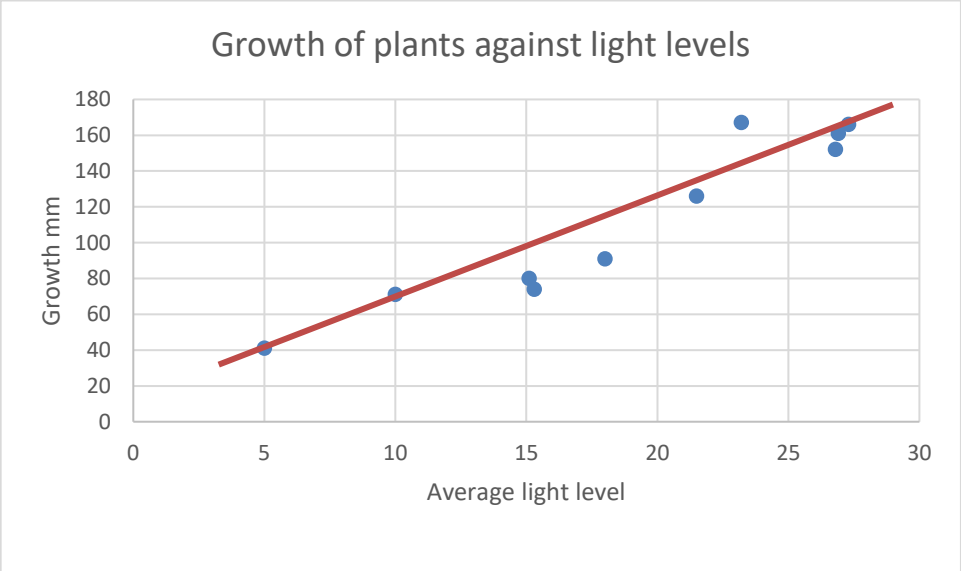
- ***Title***
- ***X axis***
- ***Y axis***
- ***2 correct points***
- ***2 additional correct points***
- ***3 additional correct points***
- ***3 additional correct points***

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Marks

b) Draw a line of best fit for your graph. **2**

Mark scheme:



1 mark each

- ***Straight line drawn in correct direction***
- ***Appropriate position of the line***

c) Explain why a pie chart would be inappropriate for the data. **1**

Mark scheme:
Pie chart requires percentages/parts of a whole (1 mark)

Total 10 Marks

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Marks

Question 2

The data for the plants is repeated here:

Plant	Average light level	Growth (mm)
1	10	71
2	15.1	80
3	23.2	167
4	26.9	161
5	27.3	166
6	18	91
7	15.3	74
8	26.8	152
9	5	41
10	21.5	126

a) Calculate *Pearson's correlation coefficient* for the given data to the nearest ONE (1) decimal place (1dp). **8**

Mark scheme:

Plant	Average light level	Growth (mm)	*xy*	x^2	y^2
1	10	71	710	100	5041
2	15.1	80	1208	228.01	6400
3	23.2	167	3874.4	538.24	27889
4	26.9	161	4330.9	723.61	25921
5	27.3	166	4531.8	745.29	27556
6	18	91	1638	324	8281
7	15.3	74	1132.2	234.09	5476
8	26.8	152	4073.6	718.24	23104
9	5	41	205	25	1681
10	21.5	126	2709	462.25	15876
	$\sum x$ = *189.1*	$\sum y$ = *1129*	$\sum xy$ = *24412.9*	$\sum x^2 = 4098.73$	$\sum y^2$ = *147225*

Award 1 mark for $\sum x$
Award 1 mark for $\sum y$
Award 1 mark for correctly completing xy ***column and*** $\sum xy$
Award 1 mark for correctly completing x^2 ***column and*** $\sum x^2$
Award 1 mark for correctly completing y^2 ***column and*** $\sum y^2$

Give credit for other appropriate workings not presented in tabular form.
Using:

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Marks

$$\boldsymbol{R} = \frac{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{\sqrt{(n\Sigma x_i^2 - (\Sigma x_i)^2)(n\Sigma y_i^2 - (\Sigma y_i)^2}}$$

$$= \frac{\mathbf{10}(24412.9) - (189.1 * 1129)}{\sqrt{(10(4098.73) - (189.1)^2)(10(147225) - (1129)^2)}}$$

$$= \frac{\mathbf{30635.1}}{32143.3769}$$

$$\boldsymbol{R} = \mathbf{0.953077}$$
$$\boldsymbol{R} = \mathbf{1.0}(\boldsymbol{1dp})$$

1 mark for correct formula completed.
1 mark for answer or r
1 mark for correct rounding of r to ONE (1) decimal place.

b) Explain what a strong positive correlation would mean in relation to the amount of light and growth. **1**

Mark scheme:
The more light the more growth (1 mark)

c) Explain what a strong negative correlation would mean in relation to the amount of light and growth. **1**

Mark scheme:
The more light the less growth (1 mark)

Total 10 Marks

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Marks

Question 3

The data for the plants is repeated here:

Plant	Average light level	Growth (mm)
1	10	71
2	15.1	80
3	23.2	167
4	26.9	161
5	27.3	166
6	18	91
7	15.3	74
8	26.8	152
9	5	41
10	21.5	126

a) Rank the results and calculate the *Spearman rank correlation coefficient* for this data. Give your answer to TWO (2) decimal places. **8**

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Marks

Mark scheme:

Plant	Average light level	Growth (mm)	Rank x	Rank y	d = rankx - ranky	d^2
1	10	71	2	2	0	0
2	15.1	80	3	4	-1	1
3	23.2	167	7	10	-3	9
4	26.9	161	9	8	1	1
5	27.3	166	10	9	1	1
6	18	91	5	5	0	0
7	15.3	74	4	3	1	1
8	26.8	152	8	7	1	1
9	5	41	1	1	0	0
10	21.5	126	6	6	0	0

$\sum \boldsymbol{d^2 = 14}$

$$\boldsymbol{r = 1 - \frac{6\sum d^2}{n(n^2 - 1)}}$$

$$\boldsymbol{r = 1 - \frac{6 \times 14}{10(10^2 - 1)}}$$

$$\boldsymbol{r = 1 - \frac{84}{990}}$$

$$\boldsymbol{r = 1 - 0.084848}$$

$$\mathbf{r = 0.915152}$$

$\boldsymbol{r = 0.92}$ ***(to 2 d.p.)***

Rank of x	***(1 mark)***
Rank of y	***(1 mark)***
Calculation of d row	***(1 mark)***
Calculation of $\mathbf{d^2}$ ***row***	***(1 mark)***
Calculation of $\sum d^2$	***(1 mark)***
Correct calculation of r	***(1 mark)***
Correct r result	***(1 mark)***
Result (to 2d.p.)	***(1 mark)***

b) State whether the data gathered by the horticulturist is qualitative or quantitative. **1**

Mark scheme:
1 mark for quantitative

c) State whether the data gathered by the horticulturist is primary or secondary. **1**

Mark scheme:
1 mark for primary

Total 10 Marks

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Marks

Question 4

The horticulturist wants to start selling plants using either an in-person store or a website and mobile app.

The horticulturist uses the TASC cycle when deciding on how to move forward with their plans.

a) Identify **THREE (3)** pieces of information that the horticulturist can gather during the *Gather/Organise* stage. **3**

Mark scheme:
1 mark each to max 3 e.g.

- ***What other websites are there?***
- ***What other local stores are there?***
- ***How much will a site/app cost?***
- ***How many people live within the area?***

b) State the purpose of the *identify* stage of the TASC cycle. **1**

Mark scheme:
Identify the criteria to assess the success by // by example applied to this scenario e.g. plant sales increase by 50%

c) In THREE (3) points explain what happens during the *decide* stage of the TASC cycle. **3**

Mark scheme:
1 mark each to max 3

- ***Positives of proposals identified/considered***
- ***Weaknesses of proposals identified/considered***
- ***Decision on what to do/open/launch is made***

d) Identify **THREE (3)** hats from the **SIX (6)** hats method that can be used during the TASC cycle. State the contribution that will be made by each hat. **3**

Mark scheme:
1 mark for each hat and contribution e.g.

- ***Black – consider weaknesses***
- ***Yellow – consider positive points***
- ***Red – consider emotional points***
- ***Blue – control/structure***
- ***White – objective/consider facts***
- ***Green – generate ideas***

Total 10 Marks

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Marks

Question 5

a) Referencing allows a reader to identify the sources of ideas and to do additional research if they need to. **3**

Identify **THREE (3)** other reasons why referencing should be used in academic work.

Mark scheme:
1 mark each to max 3 e.g.

- ***Give credit to author/source***
- ***To show that research has been completed***
- ***Avoid plagiarism***

b) Explain the difference between a reference list and a bibliography. **2**

Mark scheme:
1 mark each

- ***Reference list is all sources directly used in work***
- ***Bibliography is sources not directly used/quoted // bibliography is sources used for ideas***

c) Explain the importance of lifelong learning. Also, give **TWO (2)** examples of lifelong learning that you use and the skills each provides. **4**

Mark scheme:
1 mark each

- ***Lifelong learning definition: Learning is not only for career but should continue at all stages of life***
- ***2 examples of lifelong learning e.g. playing a sport, reading***
- ***Explanation of 1st lifelong learning e.g. skill provided (team building, critical thinking)***
- ***Explanation of 2nd lifelong learning e.g. skill provided (team building, critical thinking)***

d) Give **ONE (1)** example of continuing professional development that a computer programmer can make use of. **1**

Mark scheme:
1 mark e.g.

- ***Learn new language***
- ***Workshop***
- ***Conference***

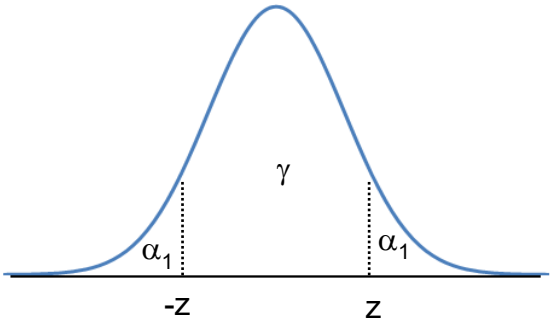
Total 10 Marks

End of paper

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Formula sheet

1. Percentage points of the normal distribution



α_1	15.87%	15%	5.00%	2.50%	2.28%	1.00%	0.50%
γ	68.27%	70.00%	90.00%	95.00%	95.45%	98.00%	99.00%
z	1.0000	1.0364	1.6449	1.9600	2.0000	2.3263	2.5758

2. Formulae

Spearman's Rank Correlation (with no ties)

$$r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

The Pearson Correlation Function

$$R = r = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{(n\sum x_i^2 - (\sum x_i)^2)(n\sum y_i^2 - (\sum y_i)^2)}}$$

Simple Linear Regression

$\hat{y} = mx_i + c$ *is the least SSE straight line where:*

$$m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \qquad m = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

$$c = \bar{y} - m\bar{x}$$

The Coefficient of Determination

$$R^2 = r^2 = \frac{\sum(\hat{y} - \bar{y})^2}{\sum(y - \bar{y})^2}$$

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Marking note

Multiply original mark out of 50 by 2 to produce final mark out of 100 to be recorded.

Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	2, 4	Yes
2	2, 4	Yes
3	2, 4	Yes
4	3	Yes
5	1, 5	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Be able to use various skills to support the study of Computing	Demonstrates little to no ability to adequately review the effectiveness and appropriateness of information and data by failing to use pre-defined techniques and/or criteria.	Demonstrates an extremely limited ability to adequately review the effectiveness and appropriateness of information and data due to insufficient use of pre-defined techniques and/or criteria.	Demonstrates a satisfactory ability to review the effectiveness and appropriateness of information and data due to adequate use of pre-defined techniques and/or criteria.	Demonstrates a very good ability to review the effectiveness and appropriateness of information and data that goes beyond the minimum required to pass due to an accurate use of pre-defined techniques and/or criteria.	Demonstrates an excellent ability to comprehensively review the effectiveness and appropriateness of information and data due to a meticulous use of pre-defined techniques and/or criteria.
Be able to communicate in a technical environment	Incorrectly identifies, adapts and makes use of a deficient range of techniques or information sources within an array of contexts.	Inconsistently identifies, adapts and makes use of a limited range of techniques or information sources within an array of contexts.	Can adequately identify, adapt and make use of a sufficient range of techniques and information sources within an array of contexts.	Can appropriately identify, adapt and make use of a range of techniques and information sources within an array of contexts with depth that goes beyond the minimum to pass.	Can consistently identify, adapt and make use of a comprehensive range of techniques and information sources within an array of contexts with depth.

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Be able to deploy thinking skills and problem-solving paradigms in both a business and learning context.	Incorrectly identifies, adapts and makes use of a deficient range of techniques or information sources within an array of contexts.	Inconsistently identifies, adapts and makes use of a limited range of techniques or information sources within an array of contexts.	Can adequately identify, adapt and make use of a sufficient range of techniques and information sources within an array of contexts.	Can appropriately identify, adapt and make use of a range of techniques and information sources within an array of contexts with depth that goes beyond the minimum to pass.	Can consistently identify, adapt and make use of a comprehensive range of techniques and information sources within an array of contexts with depth.
Be able to handle and present data	Incorrectly identifies, adapts and makes use of a deficient range of techniques or information sources within an array of contexts.	Inconsistently identifies, adapts and makes use of a limited range of techniques or information sources within an array of contexts.	Can adequately identify, adapt and make use of a sufficient range of techniques and information sources within an array of contexts.	Can appropriately identify, adapt and make use of a range of techniques and information sources within an array of contexts with depth that goes beyond the minimum to pass.	Can consistently identify, adapt and make use of a comprehensive range of techniques and information sources within an array of contexts with depth.
Understand the need for lifelong learning	Has little to no understanding of most subject knowledge and lacks an appreciation of current debates and updates in relation to this area of study.	Has insufficient understanding of most subject knowledge and has only a vague appreciation of current debates and updates in relation to this area of study.	Has an adequate understanding of some subject knowledge and demonstrates a reasonable appreciation of current debates and updates in relation to this area of study.	Has very good understanding of a wide variety of subject knowledge and demonstrates a wide appreciation of current debates and updates in relation to this area of study.	Has an extensive understanding and a comprehensive range of subject knowledge and demonstrates an insightful appreciation of current debates and updates in relation to this area of study.

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